



Pricing Review

Quotation Decision Support

- Executive Overview
- Quote Recommender
- Margin Sweet Spot
- Competitor Positioning
- Model Notes

Decision-support scope

This dashboard supports price review. It does not replace managerial approval, customer knowledge, or commercial judgement.

Version: v3.7.0

Last updated: 2026-06-12

Model: HistGradientBoostingClassifier

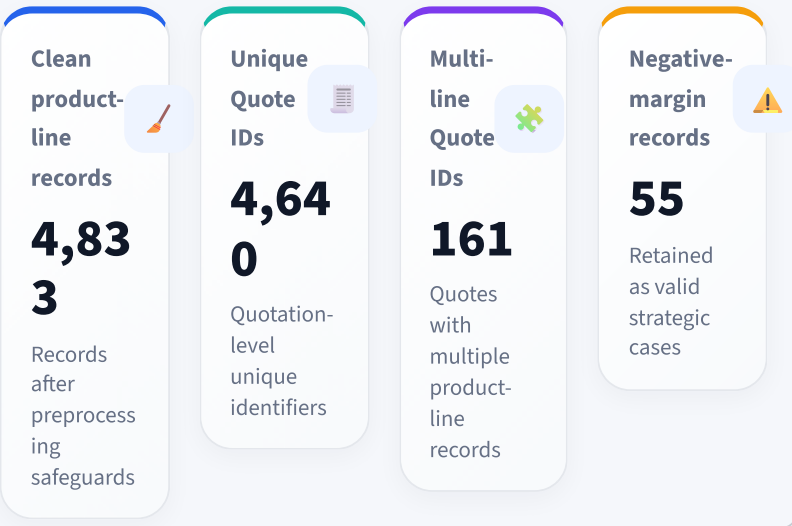
Config: initial

Quotation Pricing Decision Support Dashboard

Interactive price-review support for gross-margin management, competitor positioning, win-probability estimation, and quote-level recommendation.

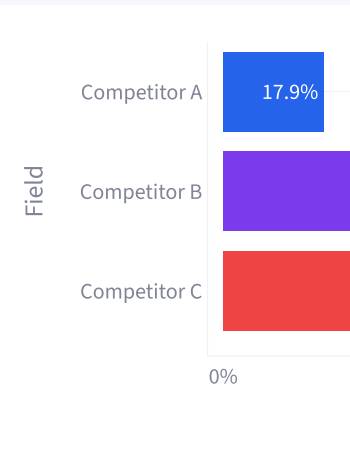
Model Notes & Data Quality

A structured overview of cleaned data, model validation results, interpretation boundaries, and data-improvement priorities.



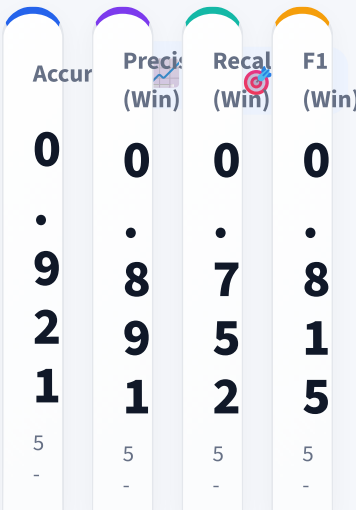
Competitor-information completeness

Historical missingness affects how competitor variables should be used operationally.



Model validation summary

Stratified 5-fold cross-validation for the final HistGradientBoosting win-probability model.



Field	Missing records	fold CV metrics	fold CV metrics	fold CV metrics	fold CV metrics
Competitor A		8	17.		
Competitor B		43	30.		
Competitor C		47	38.		

Brier Score

0.061

5-fold CV metrics

ROC-AUC

0.951

5-fold CV metrics

PR-AUC

0.903

5-fold CV metrics

Log Loss

0.220

5-fold CV metrics

Confusion matrix

Aggregated 5-fold CV classification outcomes using a 0.50 decision threshold.



Model safeguards

Safeguards applied before model training and dashboard use.

- The final website loads the saved **HistGradientBoosting** model bundle instead of training inside the app.
- The model was evaluated using **stratified 5-fold cross-validation**.
- Quantity values less than or equal

Percentages are calculated within each actual class.

to zero are excluded as system errors.

- Negative gross margins are retained as valid strategic business cases.
- Missing competitor values are not replaced with zero.
- For repeated Quote ID + Product records, the highest unit price is retained only when duplicate price variations exist.

Important interpretation notes

Boundaries for responsible interpretation of the dashboard output.

- The dashboard estimates **historical associations** between quotation features and win outcomes.
- The primary use case is **during price review**, before quotation submission.
- The recommender works at the **product-line price-review level** and aggregates results to the quote level.
- Multi-product quotations still require managerial review at the full quotation level.
- Competitor-price fields should only

Recommended next data improvements

Additional fields that would strengthen future model usefulness.

1. Add quotation timestamps and revision numbers.
2. Record whether competitor prices were known before submission or learned after the tender result.
3. Add customer segment, salesperson, region, and customer-history variables.
4. Add explicit full-quotation and product-line identifiers.
5. Record the final commercial reason for a won or lost tender when available.

be used
operationally
when available
before the
decision.

- The recommendation supports judgement; it does not automate commercial approval.

Model explanation

Global feature importance from SHAP output, if available in the asset folder.

